

PERIMETER OF IRREGULAR SHAPES

Becoming a Boundary Explorer

Session 1: The Missing Border Mystery

Can You Solve It?

Imagine you are helping your teacher decorate the classroom for a festival. You have a beautiful festival poster, but it is missing its colorful border! You have a roll of ribbon, but you don't want to waste it. How do you find out exactly how much ribbon you need to go all the way around?

Our Path Today

In this session, we are going to become "Boundary Explorers." Our goals are:

- **Define Perimeter:** Learn that it's the distance "around" the outside.
- **Explore Irregular Shapes:** Understand shapes where not all sides are equal.
- **Walk the Boundary:** Use the "Ant Walk" method to find the total distance.
- **Perimeter in Real Life:** Find examples from Rangolis to gardens!

THE QUICK QUESTION:

If you walked along the edge of your school desk, what part of the desk would you be measuring?
The top surface or the boundary?

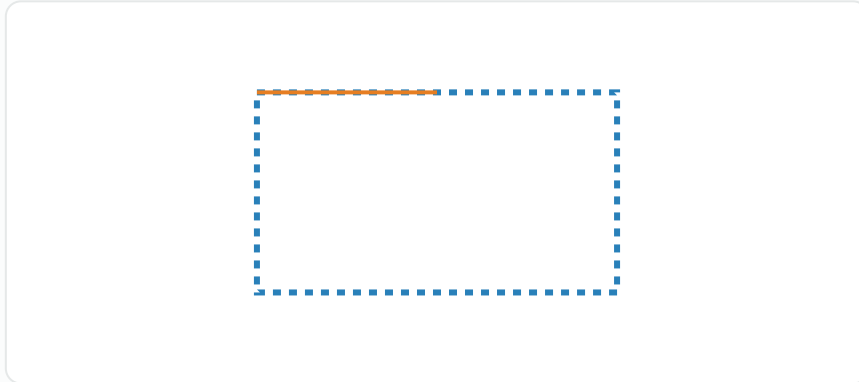
What Is Perimeter?

Perimeter is a very important concept in geometry. It tells us how long the "outline" of a shape is.

Definition: Perimeter is the total distance around the outside of a 2D shape.

The Ant Walk Mental Check

Imagine a tiny ant starting at one corner of a shape. If the ant walks exactly once all the way along the edge until it reaches the same corner again, the total distance it traveled is the Perimeter.



Perimeters Around Us in India

We use the idea of perimeter every day in India, especially during festivals and celebrations!

1. The Rangoli Border

During Diwali, when you draw a Rangoli outside your door, you might want to put a border of white powder or flower petals around it. The length of that border is the perimeter.

2. The Gota Lace Border

Have you seen your mother or grandmother putting Gota lace on a Dupatta or a Saree? They need to know the perimeter of the cloth to buy the right length of lace!

3. Colony Garden Fences

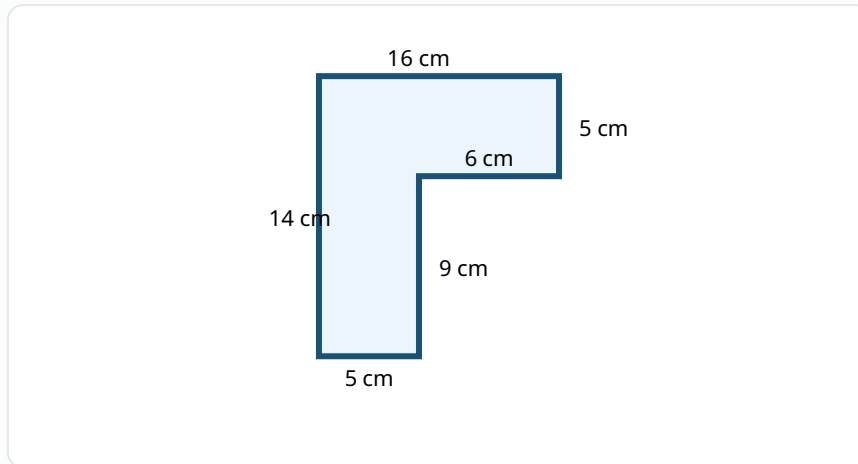
To keep the cows or dogs out of the beautiful colony park, the society puts up a fence. The length of that fence is the perimeter of the park.

Irregular Shapes: Not All Sides Are Equal!

Squares and rectangles are "regular" shapes because they follow strict rules. But many things in the real world are Irregular.

Irregular Shapes are shapes that have sides of different lengths and angles of different sizes. They don't have a standard name like "Square," but they still have a boundary!

Even if a shape looks strange or has many corners, we can still find its perimeter. We just need to be very careful to count every single side!



How to Find the Perimeter

Finding the perimeter of an irregular shape is as simple as adding!

Perimeter = Sum of ALL the side lengths

Example 1: The 4-Sided Shape

Look at a shape with these sides: 4 cm, 5 cm, 3 cm, and 6 cm. To find the perimeter, we add them up:

$$4 + 5 + 3 + 6 = 18 \text{ cm}$$

EXPLORER'S TIP: Always cross out or mark the sides you have already added so you don't count the same side twice or miss one!

Practical Activity: The L-Shaped Garden

Imagine you have a garden that is shaped like an "L". To put a stone border around it, we need the perimeter.

Step-by-Step Solution:

Let's "Walk the Boundary" for these sides: 5 cm, 7 cm, 4 cm, 6 cm, 3 cm, and 5 cm.

1. Start at the top: $5 + 7 = 12$
2. Add the next side: $12 + 4 = 16$
3. Keep going: $16 + 6 = 22$
4. And the next: $22 + 3 = 25$
5. Finally, the last side: $25 + 5 = 30$ cm

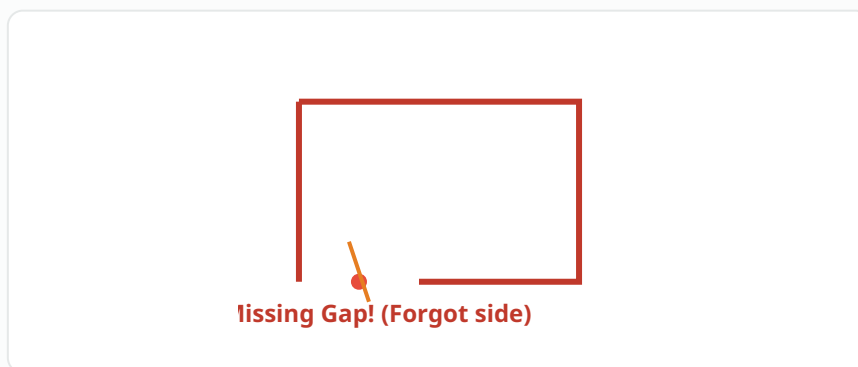
Total Perimeter = 30 cm

Why Add Every Side?

In irregular shapes, students often forget the small "turns" or "inner corners."

The Missing Gap Rule

If you forget to add even one side, your boundary will have a gap! Imagine a fence with a missing piece—the animals would get right in. To find the true perimeter, every side must form part of the sum.



Interactive Practice: Rangoli Borders

Can you find the perimeter for these two Rangoli designs?

DESIGN A (T-SHAPE)

Sides: 10 cm, 2 cm, 4 cm, 6 cm, 4 cm, 2 cm.

Your Calculation: Perimeter = _____ cm

DESIGN B (JAGGED SHAPE)

Sides: 5 cm, 5 cm, 5 cm, 5 cm, 5 cm, 5 cm, 5 cm, 5 cm.

Your Calculation: Perimeter = _____ cm

The Ant Walk Mental Check

You have successfully mastered the perimeter of irregular shapes! Let's do a final check:

- Perimeter is the total distance around a shape.
- For Irregular Shapes, not all sides are equal, but we still add them all.
- Always Walk the Boundary from start to finish.
- The total distance is the Sum of every side.

Excellent Work, Explorer!

Next time you see a Rangoli or a garden, try to "walk the boundary" in your mind!